Stress reduction with Music

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Abstract

This study investigated whether listening to music modulates physiological stress responses during acute stress induction and subsequent recovery. Two within-subject experiments used a Stroop task to induce stress, monitoring physiological responses via Heart Rate Variability (HRV) (RMSSD) and subjective stress ratings. In Experiment 1, music did not significantly enhance recovery during passive rest compared to silence. Experiment 2 found no significant difference in stress response (RMSSD) when participants listened to pink noise, their favorite song, or a disliked song during the stressor. Descriptive trends suggested higher stress during music breaks (Exp. 1) and lowest stress while listening to pink noise or disliked music during the stressor (Exp. 2). However, a very small sample size and methodological confounders limited statistical power. The study found no statistical evidence for music modulating HRV in the tested contexts, highlighting the need for future research with larger samples and stricter controls. Also, since our data was not normalized, our results could be misleading.

1 Introduction

Stress is a constant element of modern life. It arises in academic, professional, and domestic contexts and has well-documented physiological and psychological consequences. Many individuals use music as a means of relaxation, yet the effectiveness of music in reducing acute stress remains insufficiently understood. This study examines whether listening to music modulates physiological stress responses. Specifically, we compare stress induction with and without music and evaluate whether music enhances recovery during passive relaxation.

2 Methods

Two experimental protocols were conducted. Both used identical measurement procedures and the same stress-inducing task. All participants wore an electrocardiogram (ECG) from which pulse and heart-rate-variability (HRV) were derived throughout baseline, stress tests, and breaks. Participants also wore noise-cancelling headphones to eliminate ambient sound. During baseline measurements, they were instructed to close their eyes to minimize visual input.

After each baseline and each stress test, participants rated their subjective stress level using the following item on a continuous scale from 1 to 100: *"How stressed do you feel right now?"* The procedure was automated using a Python script, complemented by verbal instructions from a researcher. Our resulting data was not normalized.

2.1 Stress test

Both experiments used computerized Stroop task. Participants were shown a color word (e.g., "blue", "red") printed in an incongruent color. Their task was to indicate the displayed color by pressing a specific keyboard letter. To increase cognitive load, the letter-color mappings did not correspond to the initials of the color (e.g. blue = F, red = J, green = K, yellow = D).

Before the first stress test, participants completed a resting baseline to allow comparisons between subjects of HRV and pulse.

2.2 Experiment 1

Experiment 1 began with a 120-second baseline measurement, followed by 180 seconds of Stroop-induced stress. During the subsequent 180-second break, half of the participants listened to a predefined piece of music, while the remaining participants rested in silence.

The participants then completed a second 180-second Stroop test (in silence), followed by a second break during which the conditions were reversed (music $\leftrightarrow$ silence). This within-subject counterbalancing ensured that potential order effects did not bias the results. Table summarizes the sequence of Experiment 1.

2.3 Experiment 2

Experiment 2 began with a 150-second baseline, followed by 150 seconds of Stroop-induced stress while participants listened to pink noise. After a 120-second silent break, participants completed a second Stroop

test (150 s) while listening to their self-reported favorite song or their most disliked song.

Following another 120-second silent break, participants completed a third Stroop task while listening to the remaining song. The order of "favorite" and "disliked" songs was counterbalanced between participants. Participants had submitted their preferred and disliked song choices via a pre-experiment questionnaire. Table summarizes the schedule of Experiment 2.

Phase	**Duration**	**Task**	**Auditory Condition**	**Notes**
Baseline	120 s	Rest	Silence	Eyes closed
Stress Test 1	180 s	Stroop task	Silence	–
Break 1	180 s	Passive rest	Music / Silence	Counterbalanced
Stress Test 2	180 s	Stroop task	Silence	–
Break 2	180 s	Passive rest	Silence / Music	Counterbalanced

Table 1: Experimental schedule for Experiment 1

Phase	**Duration**	**Task**	**Auditory Condition**
Baseline	150 s	Rest	Silence
Stress Test 1	150 s	Stroop task	Pink noise
Break 1	120 s	Passive rest	Silence
Stress Test 2	150 s	Stroop task	Favourite / Disliked song
Break 2	120 s	Passive rest	Silence
Stress Test 3	150 s	Stroop task	Disliked / Favourite song

Table 2: Experimental schedule for Experiment 2

2.4 Data analysis

The analysis proceeded in four stages: data import, preprocessing, variable construction, and statistical evaluation. All steps were implemented in R using *dplyr*, *ez*, and *ggplot2*.

2.4.1 Data Import and Structure

For each participant, raw exports from the two experiments were stored in separate subdirectories. The workflow iterated through all folders, automatically detecting and loading (i) physiological summary files (`*_physio_summary.csv`) and (ii) questionnaire files (`*_questionnaires.csv`). Files belonging to Experiment 1 and Experiment 2 were identified via the folder naming convention (e.g., `5_2` → Experiment 2) and merged into two experiment-specific data frames. Data received from Experiment 1 and Experiment 2 was not normalized.

2.4.2 Physiological Data Preparation

From the physiological summaries, only the relevant phases were retained: *post_rest* for Experiment 1 and *stroop* for Experiment 2. Break-type (Experiment 1) and music-type (Experiment 2) conditions were reconstructed based on the predefined counterbalancing: for each participant, condition assignment was determined by participant parity and the temporal order of tasks (`t_start`). This yielded a clean within-subject dataset with one RMSSD value per participant per condition.

2.4.3 Statistical Analysis

For the physiological data (RMSSD), a repeated-measures ANOVA with *break_type* as the within-subject factor was applied. Additionnally, a paired Wilcoxon test was used as a non-parametric complement. Descriptive statistics were computed for each condition.

2.4.4 Visualisation

Two plot types summarised the HRV results:

1. comparative boxplots with jittered individual observations, and
2. individual trajectories with group-level mean overlays.

Both visualisations were generated separately for Experiment 1 and Experiment 2.

3 Results

3.1 Test 1

The RMSSD values show only small descriptive differences between the two break conditions. Mean and median values are similar across groups (Music: 58.2 ms vs. Silence: 63.6 ms), and both conditions exhibit substantial inter-individual variability, reflected in the large standard deviations and wide IQRs. Overall, the descriptive statistics do not indicate a systematic HRV increase or decrease associated with either music or silence. See table 3.

The repeated-measures ANOVA did not reveal a significant effect of break type on RMSSD. The generalized effect size was very small ($\eta^2_G = 0.010$), indicating that only a negligible proportion of the variance in HRV can be attributed to the difference between music and silence. Overall, the ANOVA provides no statistical evidence for a condition effect in Experiment 1. See table 4. In figure 1 the general trend is, that during stress depletion, RMSSD is lower with music than without music.

Break Type	n	Mean	SD	Median	Min	Max	Q1	Q3	IQR
Music	6	58.2	26.7	60.7	28.9	100.0	36.4	68.1	31.7
Silence	6	63.6	32.0	60.8	31.3	114.0	37.6	78.5	41.0

Table 3: Descriptive statistics of RMSSD values (Experiment 1).

Effect	DFn	DFd	F	p	η^2_G
Break Type	1	5	2.79	0.156	0.010

Table 4: Repeated-measures ANOVA on RMSSD (Experiment 1).

3.2 Test 2

The RMSSD values differ only modestly across the three listening conditions. Mean and median values for favourite song, hate song, and pink noise are broadly comparable (46.4–52.8 ms), and all conditions show substantial inter-individual variability, as indicated by the large standard deviations and wide ranges. IQRs are consistently narrow (13.6–19.3 ms), suggesting similarly heterogeneous responses across the three auditory stimuli. Overall, the descriptive statistics do not indicate a systematic RMSSD increase or decrease uniquely attributable to any of the three music types. See Table 5. In figure 2, a trend can be observed. The RMSSD is lower when listening to the favorite song and higher when listening to the hated song or pink noise

Music Type	n	Mean	SD	Median	Min	Max	Q1	Q3	IQR
Favourite Song	5	46.4	29.2	40.7	9.87	89.5	36.5	55.6	19.1
Hate Song	5	49.9	27.9	48.2	11.2	87.7	41.7	60.9	19.3
Pink Noise	5	52.8	33.9	48.6	7.36	102.0	46.3	59.9	13.6

Table 5: Descriptive statistics of RMSSD values (Experiment 2).

Effect	DFn	DFd	F	p	η^2_G
Music Type	2	8	2.65	0.131	0.009

Table 6: Repeated-measures ANOVA on RMSSD (Experiment 2).

The repeated-measures ANOVA did not reveal a significant effect of music type on RMSSD. The observed effect size was extremely small ($\eta^2_G = 0.009$), indicating that only a negligible proportion of the variance is explained by differences between favourite song, hate song, and pink noise. Mauchly's test showed no violation of sphericity ($W = 0.891$, $p = 0.841$), and both Greenhouse–Geisser and Huynh–Feldt corrections yielded the same non-significant conclusion. Overall, the ANOVA provides no statistical evidence for a condition effect in Experiment 2. See Table 6.

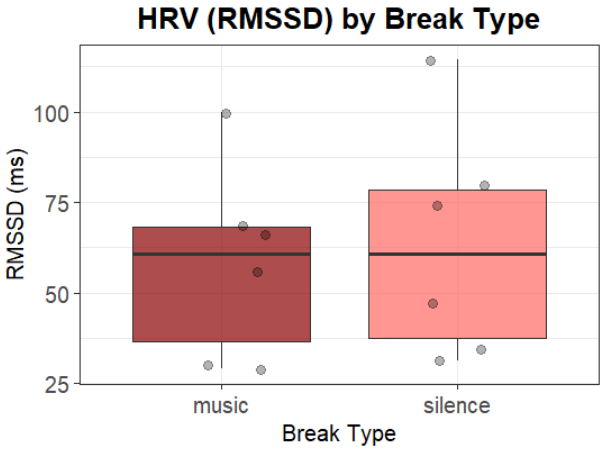


Figure 1: HRV by break type (experiment 1).

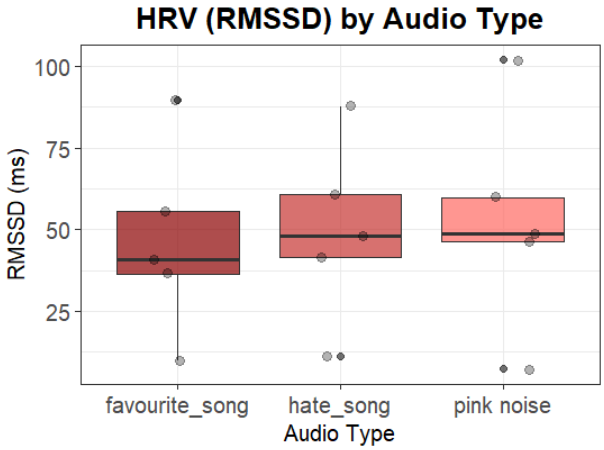


Figure 2: HRV by audio type (experiment 2).

4 Discussion

The observed effects were not statistically significant or normalized, but two descriptive trends emerged: (1) stress levels tended to be higher during breaks when music was played, and (2) during stress induction, the highest stress occurred when participants listened to their favorite song, whereas pink noise and disliked music were associated with lower stress values.

These patterns only partially align with existing literature. Burns et al., 2002 reported that music influences the cognitive component of the stress response. However, Fallon et al., 2020 found the opposite in a comparable design, showing greater stress reduction during breaks in the music condition compared to silence. With respect to experiment 2, Labbé et al., 2007 showed that self-selected (and thus generally preferred) music reduces negative affect and physiological arousal compared to silence, whereas Song et al., 2024 reported no evidence that music facilitates physiological recovery after a standardized stressor. These inconsistencies in prior research highlight that the effect of music on stress depends strongly on context, methodology, and the type of music used.

The sample size in the present study was too small to reliably test the hypotheses or to fully counterbalance possible order effects. As a result, individual differences may have disproportionately influenced mean values and variance.

Several methodological factors may also have contributed to measurement imprecision:

- **Caffeine intake.** Coffee consumption prior to testing can elevate stress and reduce HRV, introducing uncontrolled variability.
- **Colour blindness.** The Stroop test relies on accurate colour discrimination. Two participants reported colour blindness, which may have altered task difficulty and thus the stress response.
- **HRV measurement accuracy.** HRV derived from optical sensors is known to be less precise than HRV obtained via ECG. Strap positioning and tightness can further worsen signal quality.
- **Environmental variability.** Although all sessions took place in the same room, testing times varied between participants, and minor differences in instructions or interactions may have influenced stress levels.
- **Familiarity effects.** Participants with prior experience in Stroop or reaction-time tasks may respond differently to the stressor than novices.
- **Learning across experimental blocks.** Because all participants completed both experiments, they may have been more familiar with the tasks during the second run, reducing stress induction and affecting HRV.

Overall, the combination of a small sample size and several uncontrolled confounders limits the interpretability of the results. Future studies should increase sample size, ensure stricter control of participant preparation and measurement conditions, and consider using ECG-based HRV assessment to improve accuracy.

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